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PI Name:	Natarajan, Mohan Ph.D.		
Project Title:	Targeting Protein Cross Talk Signaling as a Biological Predictor for Space Radiation Cancer Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Task Description:	One of the important concerns of 'late effects' after space radiation exposure is the onset of cancers. The onset of cancer could be initiated by radiation-induced molecular changes that are persistently maintained through a positive feed back cycle and amplified over time. We propose that two key proteins, nuclear factor kappa Beta (NF-kB) and tumor necrosis factor alpha (TNF-a), sequentially co-ordinate and control this cascade of events. A novel role of these proteins in radiation-induced tumor development is identified through our recent in vitro and in vivo studies after low LET radiation exposure. In this project we address: (i) how low doses of high LET radiation trigger initial damage signals that remain persistent and are maintained in the memory of the cell/tissue, (ii) how these signals are amplified, (iii) how these amplified signals are transmitted to result in a sustained radial expansion of initial damage, and finally (iv) how radial communication of these signals activates factors that are responsible for predisposing the susceptible cells to carcinogenesis. We will examine these objectives using both in vitro (parallel plate flow model) and ex-vivo (mouse-artery organ culture) approaches, in which irradiated normal vascular endothelial cells in an intact tissue setup or organ culture will be allowed to communicate with the co-existing non-targeted normal or pro-tumorigenic cells from lung, skin, and breast tissue. An in vivo animal model will be used to prove the tumorigenic potentials of transformed cells. We will compare the outcomes with regard to the LETs (51 and 181 keV/um) of two different HZE ion beams (28Si and 56Fe) accelerated to the same velocity (600 MeV/amu) and having similar track structure dimensions, but different ionization densities. This project addresses four of the seven major risks and gaps (Cancer 1, 2, 3, & 4) identified in the Space Radiation risks.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This study will clarify whether radiation-induced molecular changes that are persistently maintained through positive feed-back cycle through two key proteins, nuclear factor kappa B (NF-kB) and tumor necrosis factor alpha (TNF-a) are amplified over time. These altered molecular events are responsible for clonal expansion of cells that are prone to be tumorigenic. In terms of usefulness of the above study on Earth, normal tissue response is one of the major focuses to achieve acceptable therapeutic gain in radiotherapy of many cancers. Identification and characterization of the molecular mediators that play a role in radiation-induced bystander effects that could lead to genomic instability and survival advantage after therapeutic doses of low LET radiation is important to understand the radiation-induced delayed effect in normal tissue. Therefore the findings from the study proposed with space radiation can set a stage to understand the underlying mechanism of tumorigenesis, tumor recurrence and tumor growth at the treatment site.		
	Alterations in the architectural network of the cellular signaling mechanism by modifying the functional aspects of signaling mediators and fixation of those alterations may translate into oncogenic processes leading to carcinogenesis. However, it is not yet known how these mediators are orchestrated to result in the induction of cancer. In cases such as low fluxes of heavy ion radiation, where only less irradiated cells (primarily at low fluences) co-exist with more nonirradiated counter parts, the alterations may be predominantly through an intercellular communication between irradiated to non-targeted cells than merely through the irradiated cells themselves. Therefore non-targeted effect (bystander effect) is considered to be one of the main factors for cancer initiation. Second, it is highly unlikely that an independent factor, as reported in most of the radiation-induced bystander studies, can provide answers to alleviate cancer risk uncertainties. Unless the signals are recycled and amplified, it is possible that an intact homeostatic mechanism may recognize and dissipate these signals, This concept was tested in human aortic endothelial cells exposed to two different HZE ion beams (28Si and 56Fe) accelerated to the same velocity (600 MeV/amu) and having similar track structure dimensions, but different ionization densities. For comparison, gamma radiation (137Cs) at a total dose of 0.8 or an equidose of 1.6 Gy were used. In this second year funding period we performed and completed several experimental approaches proposed in specific aim 1 and aim 2. The summary of the experimental results are presented below: Human aortic endothelial cells exposed to iron ion (56Fe/600 MeV/amu) to a total dose of 0.1, 0.2, 0.4, or 0.8 Gy were harvested after different post exposure time. Nuclear extracts prepared from these samples as described in our earlier publications were analyzed for the activation of nuclear factor kappa B (NF-kB) by electrophoretic mobility shift assay (EMSA). NF-kB activation was seen in cells exposed at a dose as low as 0.1 Gy of iron ion compared to mock-irradiated cells. Similar expression levels were seen after 0.2, 0.4, and 0.8 Gy. There was a slight increase in the activation of NF-kB with increasing		

Task Progress:	dose. A maximum activation in all doses was observes between 6 and 12 h. Next, human aortic endothelial cells exposed to 0.8 Gy of iron ion (56Fe; 600 MeV/ne) and harvested after 15 min, 3, 6, 12, 24, 36, 48, and 72 h showed a time-dependent NF-kB DNA binding activity, which persists even after 72 hours, the longest time point examined in this study. Quantitative analysis showed an increase in NF-kB activity at 6 h and reduced by 50% at 12, increased again at 24 h. The activation is again reduced at 36 and 48 h and then started increasing at 72 h. The results indicated a three phase activation of NF-kB after heavy ion exposure suggesting an active positive feedback cycle through autocrine signaling mechanism. Second, we developed IKK null animal model to understand the involvement of NF-kB signaling in the bystander processes that could lead to cancer onset after heavy ion exposures. The Ikk floxed mice were bread and litter-mates at 3 weeks old were genotyped with IKK f/f forward [5' CCT TGT CCT ATA GAA GCA CAA C 3'] and reversed [5' GTC ATT TCC ACA GCC CTG TGA 3'] primers, respectively. The vascular endothelial cells were isolated from abdominal and thoracic aortae of those mice that were homozygous for IKK flox (Ikk+/+). Isolated cells were propagated in M1168 cell culture medium containing 10 % serum (CellBiologicals, Chicago, IL). The subcultured cells at passage 4 were transfected (1ug/6 cm plate) with GFPcre fusion construct with a strong EF1 promoter (pBS598 EF1alpha-EGFPcre; Addgene, Cambridge, MA) by electroporation using Neon transfection system (Life Technologies, CA) at a pulse rate of 2 (n), pulse voltage of 1400 (V), and pulse width of 20 (ms). GFP expression examined by FACS analysis and confocal microscopy validated the transfection efficiency. Minimal or complete absence of NF-kB DNA-binding shown by EMSA analysis in the nuclear extracts of cre-EGFP transfected IKKf+/+ endothelial cells confirmed the loss of IKK/NF-kB activity. These genetically modified cells will be exposed to 56Fe or 28Si to prove the autocrine and paracrine bystander effect triggered by heavy ion radiation is mediated through NF-kB signaling cascade. Third, we have demonstrated that sustained activation of NFkB caused by low LET radiation initiated TNFa-NFkB feedback signaling leads to radioresistance and tumor recurrence. As an extension of this study we investigated whether or not function-specific orchestration of NFkB in response to radiation reveals a molecular blue-print that dictates induced tumor relapse and metastasis. We investigated whether muting IR-triggered or TNFa-dependent second-signaling feedback-dependent NFkB nuclear import results in limiting IR-altered invasion and metastasis. Cells were exposed to 2 Gy and incubated for 1 h or 24 h. The cells were then treated with an NFkB-targeting peptide blocker, SN50. Upon confirming the blockade in DNA-binding activity, transcription driven transactivation of NFkB and secretion of soluble TNFa, transcriptional alterations of 93 tumor invasion/metastasis genes were assessed by using QPCR profiling and then were selectively validated at the protein level. Exposure to 2 Gy induced 63, 42, and 71 genes in surviving SH-SY5Y, IMR-32, and SK-N-MC cells, respectively. Blocking post-translational nuclear import of NFkB comprehensively inhibited both initial activation of genes (62/63, 34/42, and 65/71) triggered by IR and also TNFa-mediated second signaling-dependent sustained (59/63, 32/42, and 71/71) activation of tumor invasion and metastasis signaling molecules. Furthermore, alterations in the proteins MMP9, MMP2, PYK-2, SPA-1, Dnmt3b, Ask-1, CTGF, MMP10, MTA-2, NF-2, E-Cadherin, TIMP-2, and ADAMTS1 and the results of our scratch-wound assay validate the role of post-translational NFkB in IR-regulated invasion/metastasis. These data demonstrate that IR-induced second-phase (post-translational) NFkB activation mediates TNFa-dependent second signaling and further implies that IR induced NFkB in cells that survive after treatment could regulates tumor invasion/metastasis signaling. This proof of concept is being examined with 56Fe at 600MeV.amu at doses 0.4 and 0.8 Gy at a dose rate of 0.5 Gy/min. Fourth, a flow shear system was first developed at UTHSCSA for low LET exposure. The results out of the low LET exposure were discussed in the previous year annual report. In the current year a similar flow shear system was established for high LET radiation exposure at the NSRL facility. Mechanistic approach, in general, studied in detail in in vitro cell culture experiments. However, the two-dimensional in vitro cultures may limit the understanding of the combined effect of other supporting factors that occur in in vivo micro- and macro-environment. Therefore, it is important to understand the contributions of supporting factors such as extracellular matrix and local environment in which the cells/tissue reside; in particular, with endothelial cells, where the cells are constantly undergoing different types of shear stress due to continuous blood flow. Considering these issues, in this study, we propose to investigate the high LET radiation– induced signaling responses in primary human aortic endothelial cells grown on matri-gel (simulated extra cellular matrix)-coated surface and experiencing physiological hemodynamic flow shear stress (that occurs in normal blood vessels). In addition, using this parallel plate culture, instead of an in vivo model, identification of the role of a gene or gene product in a specific pathway by blocking experiments can be studied. The beam shape, beam uniformity and staging were optimized. Using this system, primary endothelial cells were exposed to 56Fe or 28Si at 600MeV/amu at a total dose of 0.2, 0.4, and 0.8 Gy. One set of cells were harvested to examine NF-kB activation and another set of cells were collected to determine the levels of inflammatory markers. EMSA analysis showed an increased NF-kB activation in cells that are subjected to shear stress (16 dynes/cm2) prior to exposure to radiation compared to cells that are in static culture and then exposed to radiation. Further post-exposure time points are being examined to determine the differential activation of NF-kB-dependent signaling pathway and the extent of the influence of microenvironmental niche. Finally, we investigated dinuclear and heptanuclear complexes of copper (II) with 7-azaindole ligand in order to develop a compound for radio resistant counter measure. The initial inorganic characterization of this compound has been validated and published. Further experiments to understand the potentials of this compound in radio resistance is being investigated. The ongoing experiments proposed in aim 3 on in vivo animal model exposed to 56Fe and 28Si are anticipated to be completed in the next four to six months. Manuscripts that are in preparation (1. Protein cross-talk signaling regulates radiation-induced bystander effects and initiates cancer pre-disposing determinants; and 2. Influence of hemodynamic flow shear stress in differential NF-kB dependent gene expression after space radiation exposure) will be submitted in the next three months.
Bibliography Type:	Description: (Last Updated: 04/11/2021)
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